Matrix functions and stability

Preservation of real H-matrix structure by Newton square-root iterates

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Rating rationale: Challenging because convergence does not control comparison-matrix structure at each iterate; specialist impact concerns square-root algorithms for H-matrices.

Resolution — 2026-09-11

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Every exact Newton square-root iterate initialized at $X_0 = A$ remains a real nonsingular H-matrix with positive diagonal. The theorem includes arbitrary positive scalar scaling and nonnegative affine initializations, with one diagonal-dominance weight for all iterates; it also proves the corresponding Halley preservation result.

The complete target is resolved. Its former difficulty rating is historical; the original statement, references and dated audits remain below.

Primary reference: complete authored PDF, standalone TeX, **Theorem 1; Corollary 5 gives the Halley extension**. Independent proof review · Authorship and submission record. Verification is independent agent review, not external human peer review or formal certification.

Problem statement

For a real square matrix $Y = (y_{ij})$, define its comparison matrix by $\mathcal{M}(Y)_{ii} = |y_{ii}|$ and $\mathcal{M}(Y)_{ij} = -|y_{ij}|$ for $i \neq j$. Call Y a nonsingular H-matrix when $\mathcal{M}(Y) = sI - B$ for some entrywise nonnegative B and real $s > \rho(B)$.

Let $n \geq 1$ and let $A \in \mathbb{R}^{n\times n}$ be a nonsingular H-matrix with $a_{ii} > 0$ for every i. Consider the exact-arithmetic recurrence

$$X_0 = A, \qquad X_{k+1} = \tfrac{1}{2}(X_k + X_k^{-1}A), \qquad k \geq 0.$$

Is every X_k a nonsingular H-matrix with strictly positive diagonal? The known existence of these iterates and their convergence to the principal square root do not themselves establish the requested structure at every step.

References and status

N. J. Higham, *Functions of Matrices: Theory and Computation* (SIAM, 2008), §6.3, equation (6.12), §6.8.3, and Research Problem 6.25, p. 170. The author's errata correct the surrounding p. 161 theorem to real H-matrices; that restriction is adopted here. L. Lin and Z.-Y. Liu, *On the Square Root of an H-matrix with Positive Diagonal Elements*, *Annals of Operations Research* 103 (2001), 339–350, concern the square root itself. Searches on 2026-09-08 for the problem number and Newton/H-matrix structure preservation found no resolution. The recent Bini–Iannazzo–Meini–Meng preprint (20 May 2026), §§3–4, treats M-matrices, a narrower class. The latest explicit open-problem evidence found for this assertion remains the 2008 book.

Audit — 2026-09-10

Rechecked the 2026 M-matrix paper, §§4 and 4.2: its normalized nonsingular M-matrix class has structure-preserving Newton iterates starting from A. This is a substantive subfamily of the displayed target. Searches for Higham's Problem 6.25 and H-matrix iterate preservation found no general answer; the surviving H-matrix question still has historical evidence.